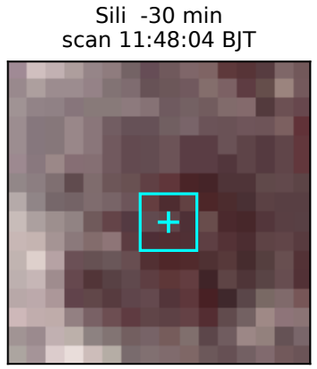
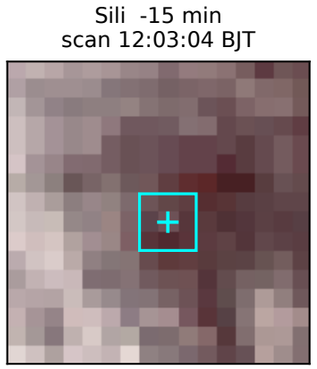
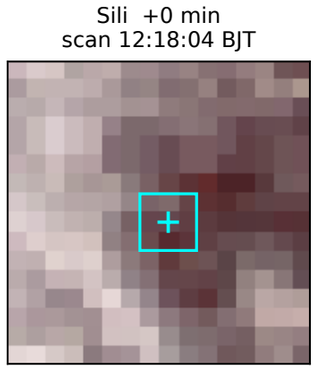
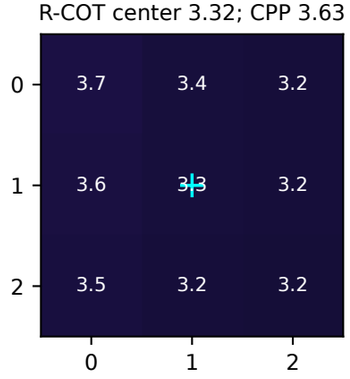
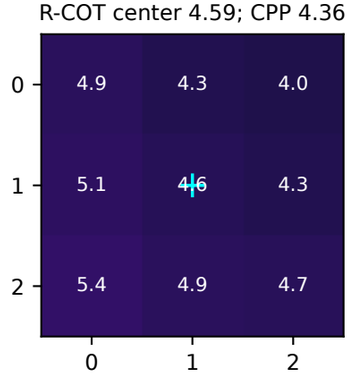
Paired true-cloud forensics 1/6 · fixed 3×3, three-frame GHI models

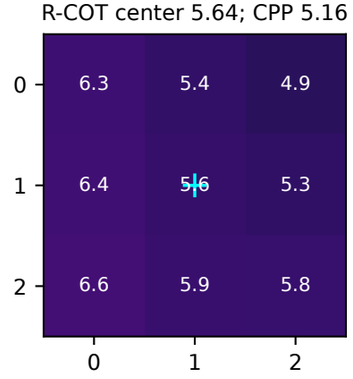


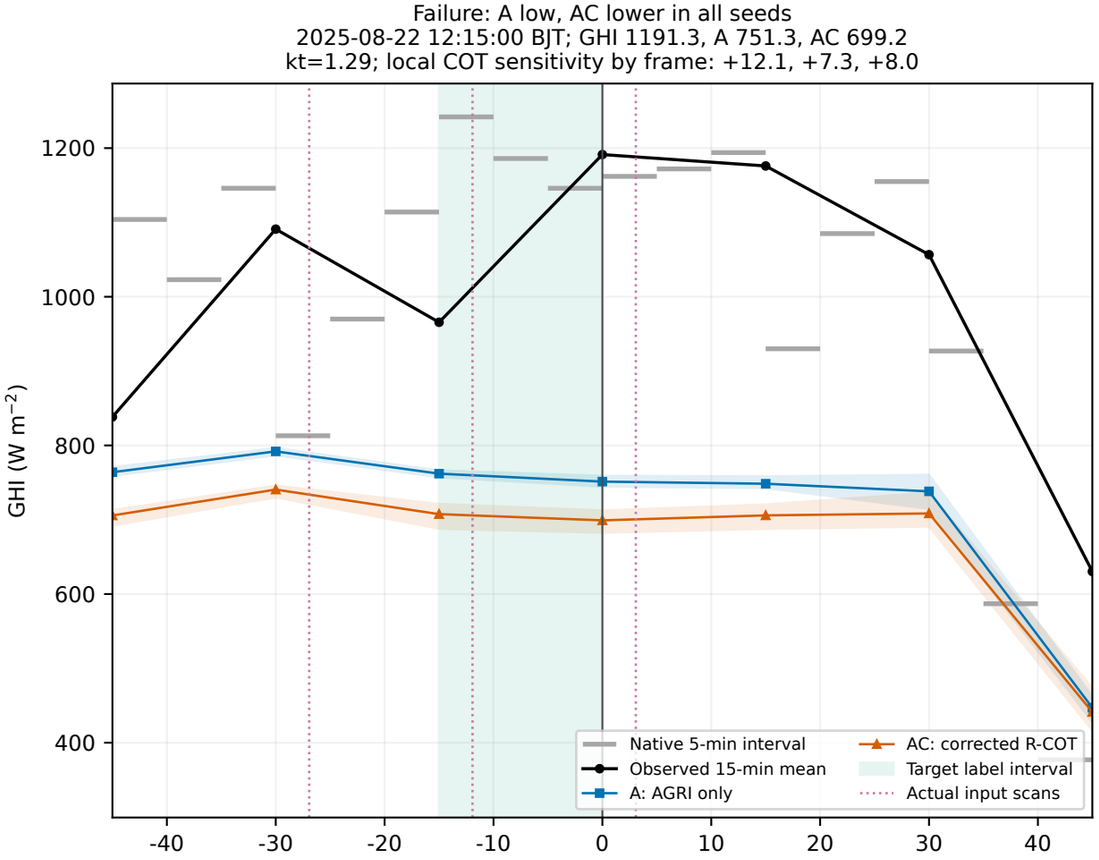


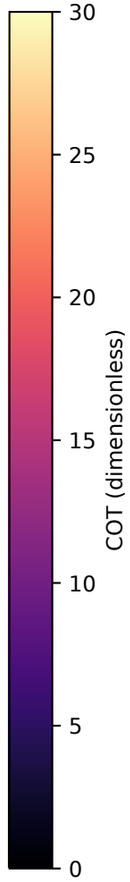


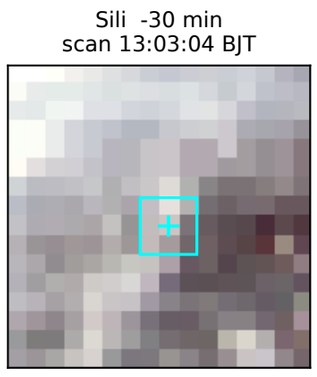


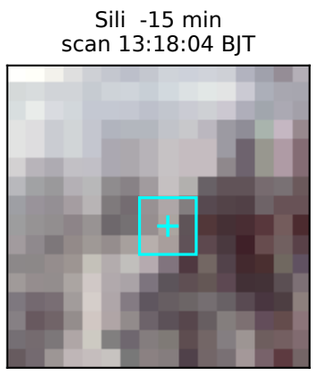


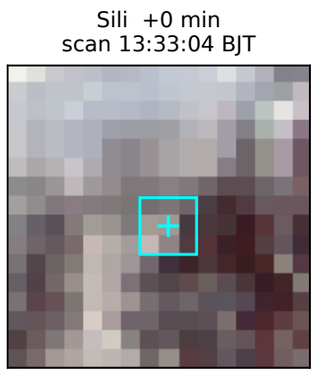


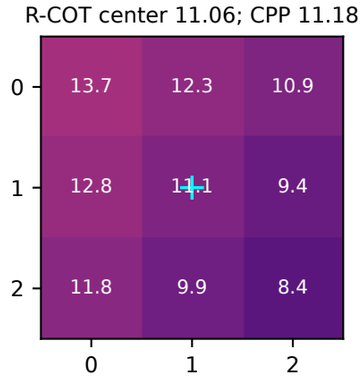


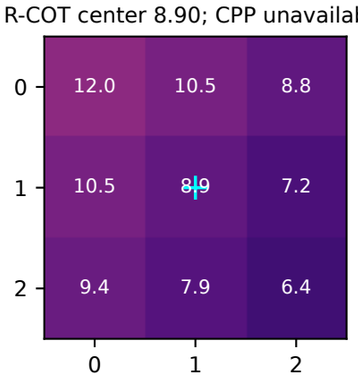


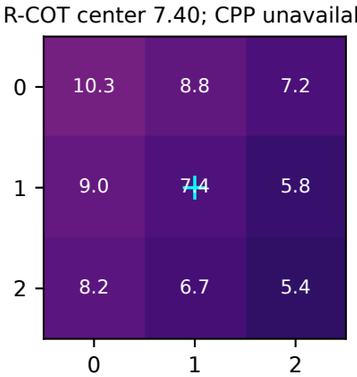


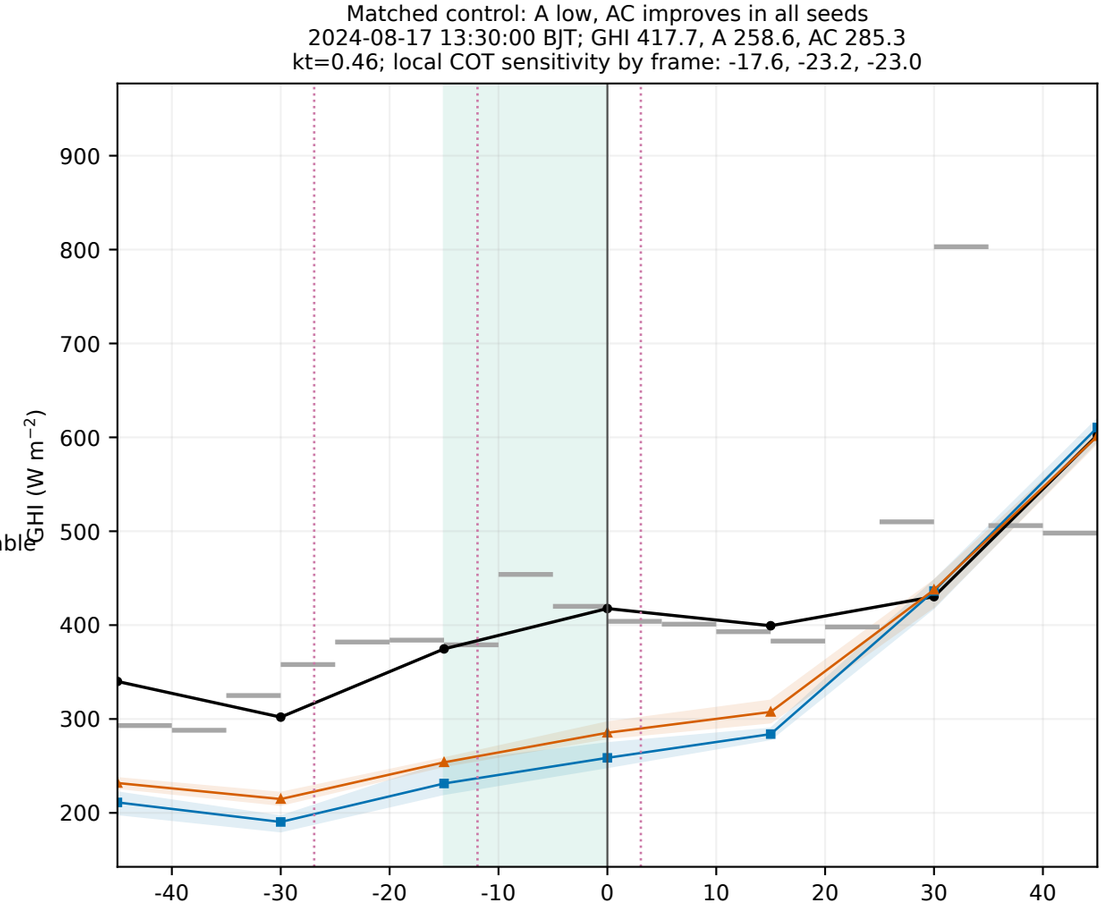












RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

Paired true-cloud forensics 2/6 · fixed 3×3, three-frame GHI models

Sili -30 min
scan 16:18:04 BJT



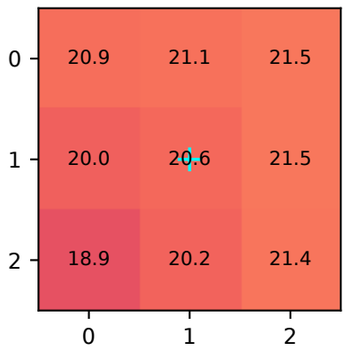
Sili -15 min
scan 16:33:04 BJT



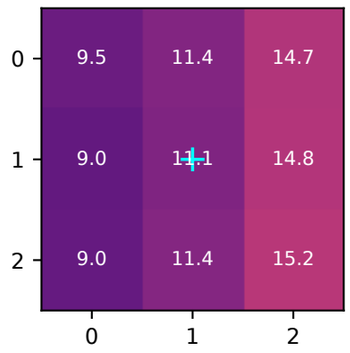
Sili +0 min
scan 16:48:04 BJT



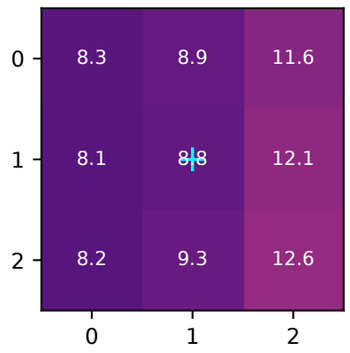
R-COT center 20.58; CPP 26.20



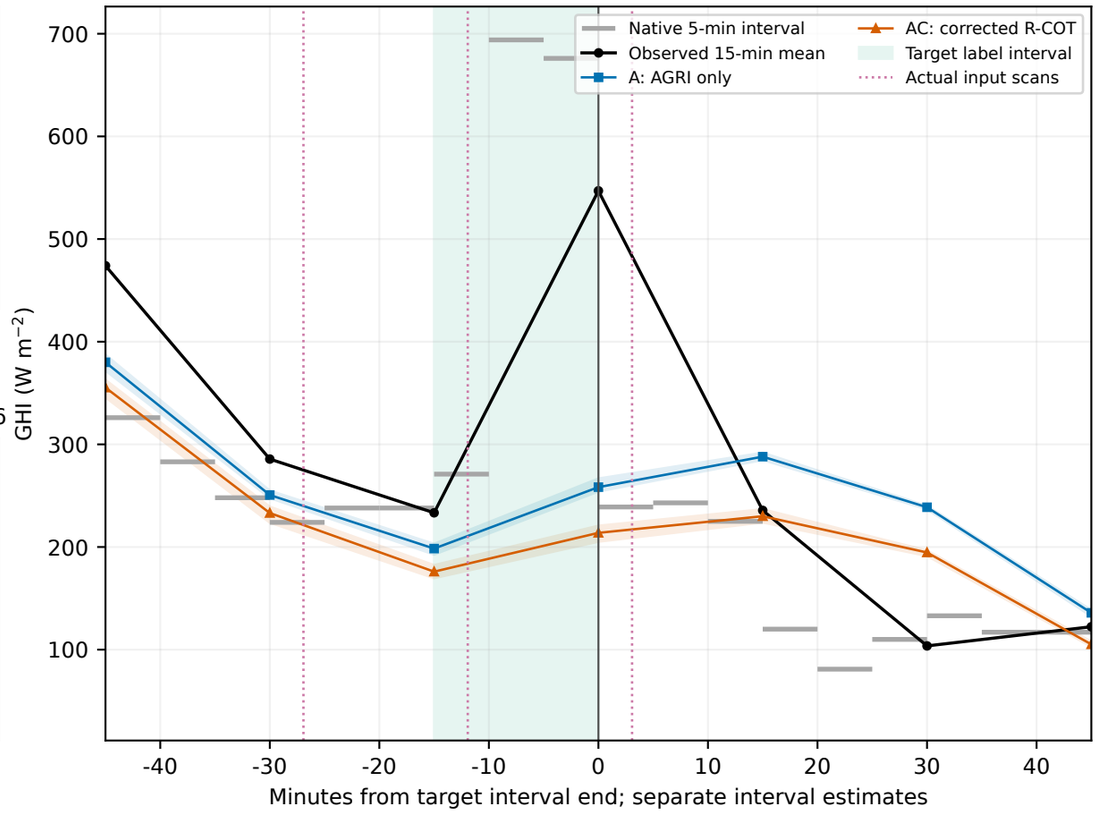
R-COT center 11.08; CPP 12.86



R-COT center 8.79; CPP 11.16



Failure: A low, AC lower in all seeds
2025-08-21 16:45:00 BJT; GHI 547.0, A 258.3, AC 213.8
kt=1.30; local COT sensitivity by frame: -33.5, -34.5, -31.8



Sili -30 min
scan 16:33:04 BJT



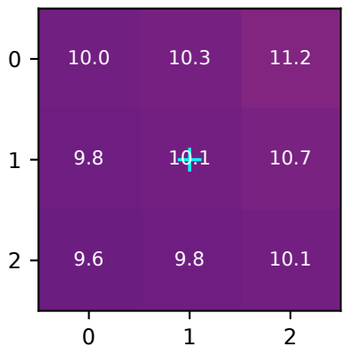
Sili -15 min
scan 16:48:04 BJT



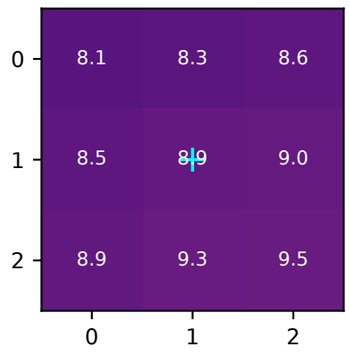
Sili +0 min
scan 17:03:04 BJT



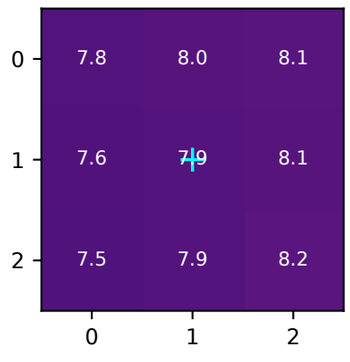
R-COT center 10.11; CPP unavailable



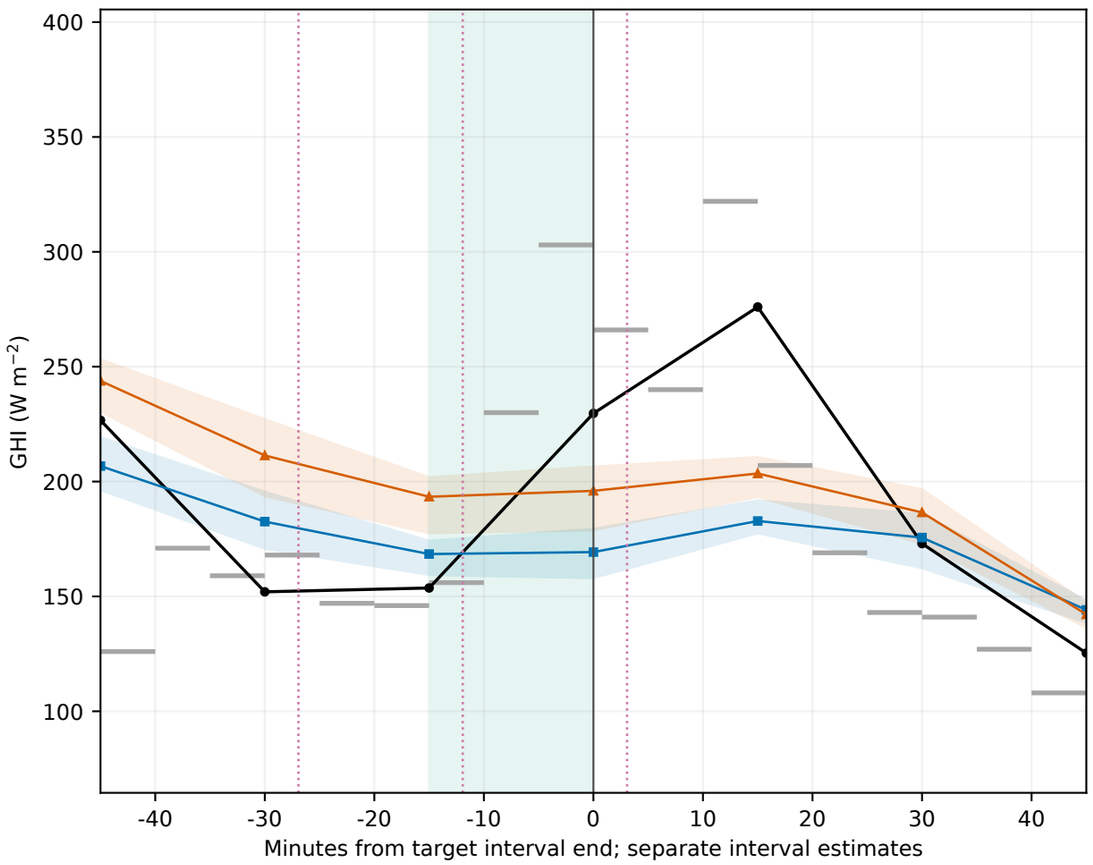
R-COT center 8.93; CPP unavailable



R-COT center 7.92; CPP 8.26

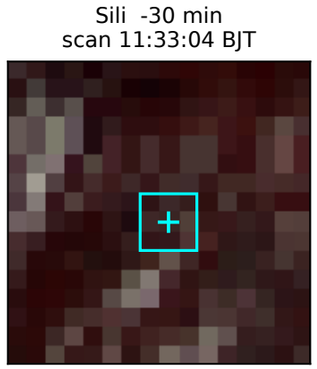


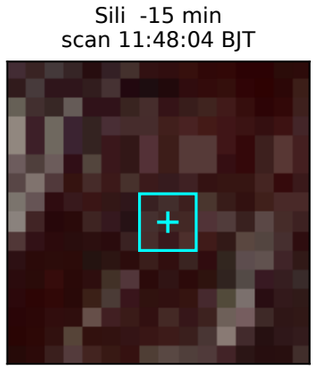
Matched control: A low, AC improves in all seeds
2024-06-12 17:00:00 BJT; GHI 229.7, A 169.3, AC 195.9
kt=0.56; local COT sensitivity by frame: -20.3, -21.7, -19.9



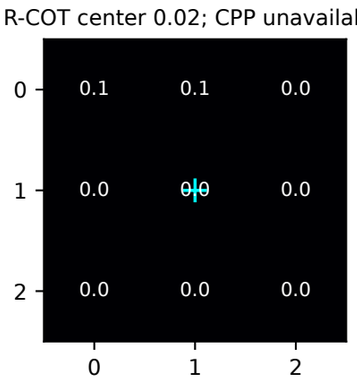
RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

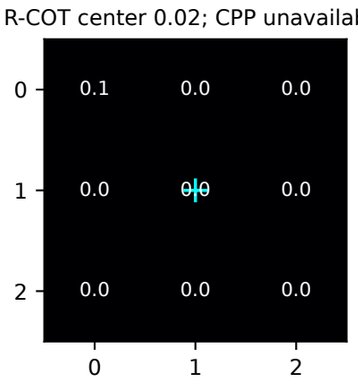
Paired true-cloud forensics 3/6 · fixed 3×3, three-frame GHI models

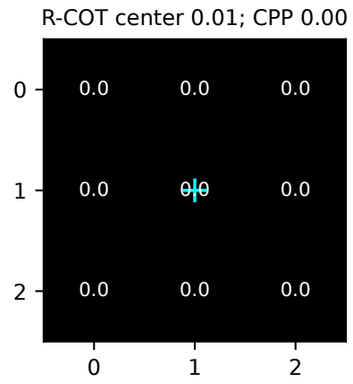


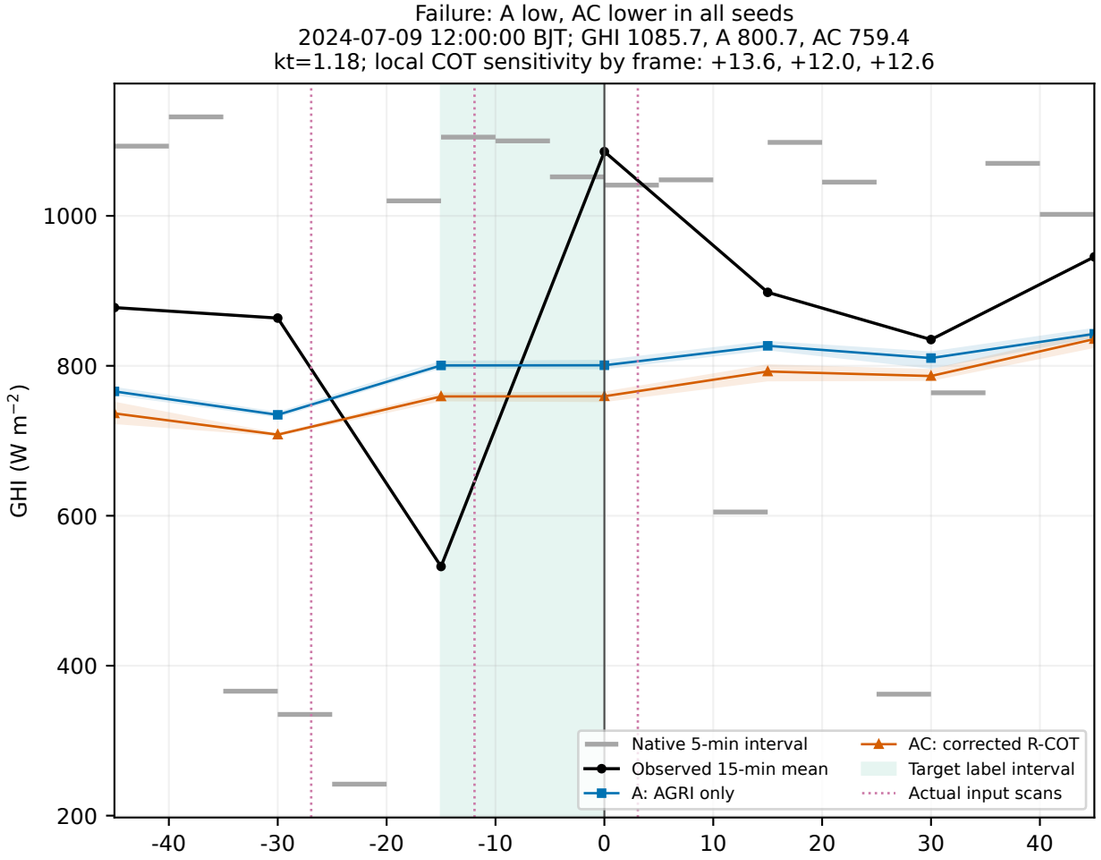


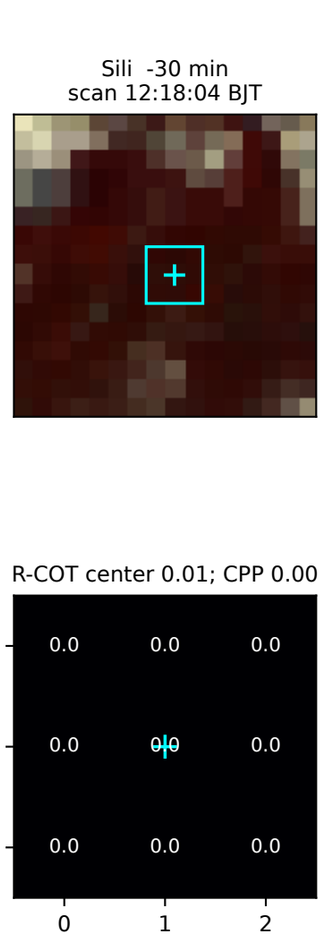


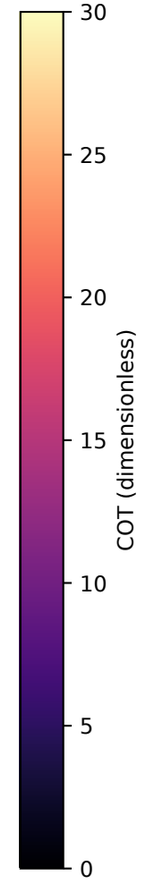


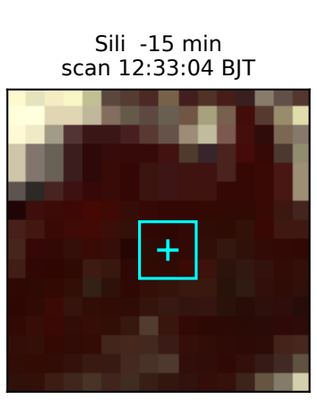


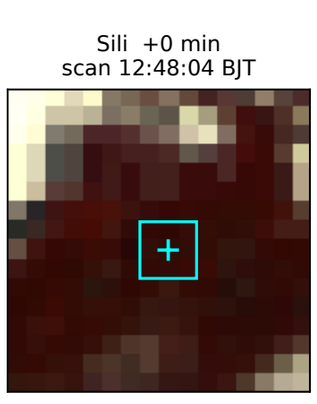


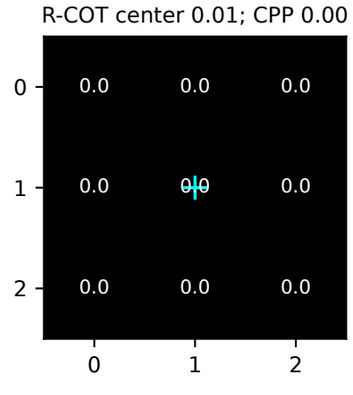


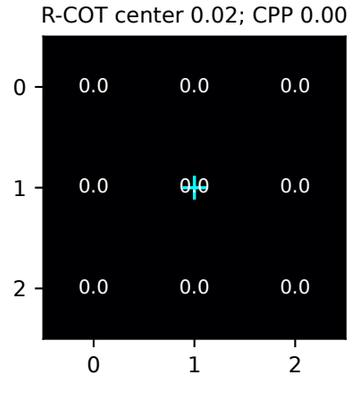


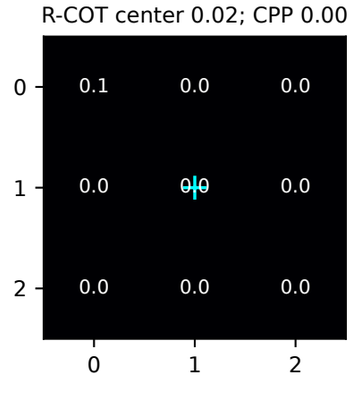


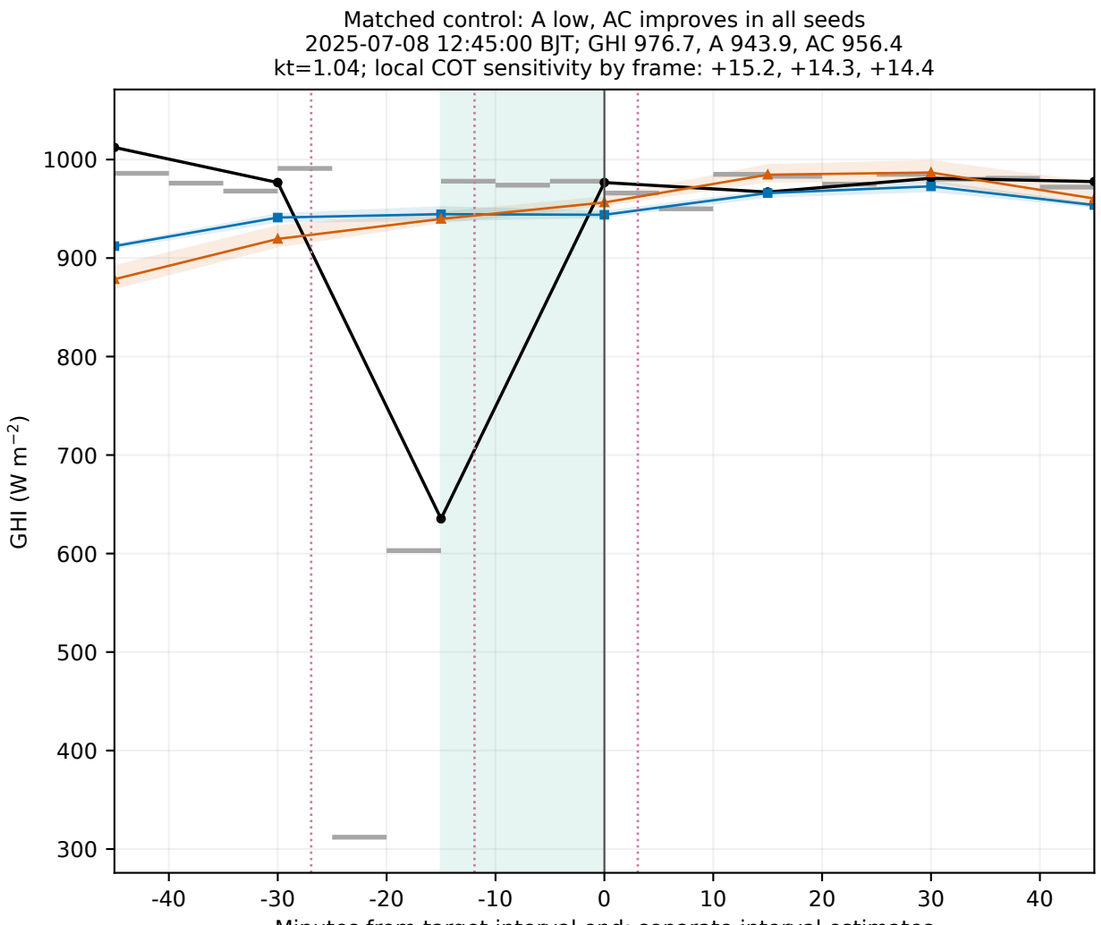












RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

Paired true-cloud forensics 4/6 · fixed 3×3, three-frame GHI models

Zhujia -30 min
scan 10:17:39 BJT



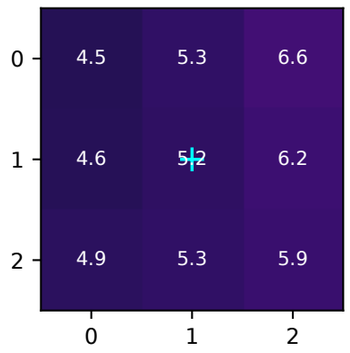
Zhujia -15 min
scan 10:32:39 BJT



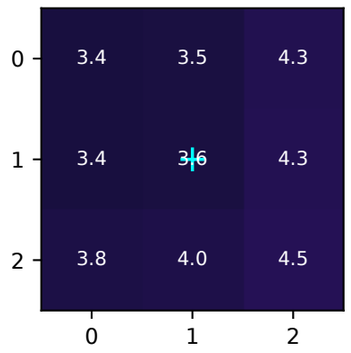
Zhujia +0 min
scan 10:47:39 BJT



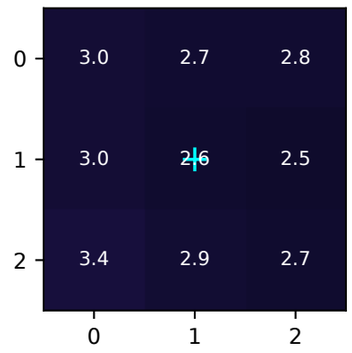
R-COT center 5.23; CPP unavailable



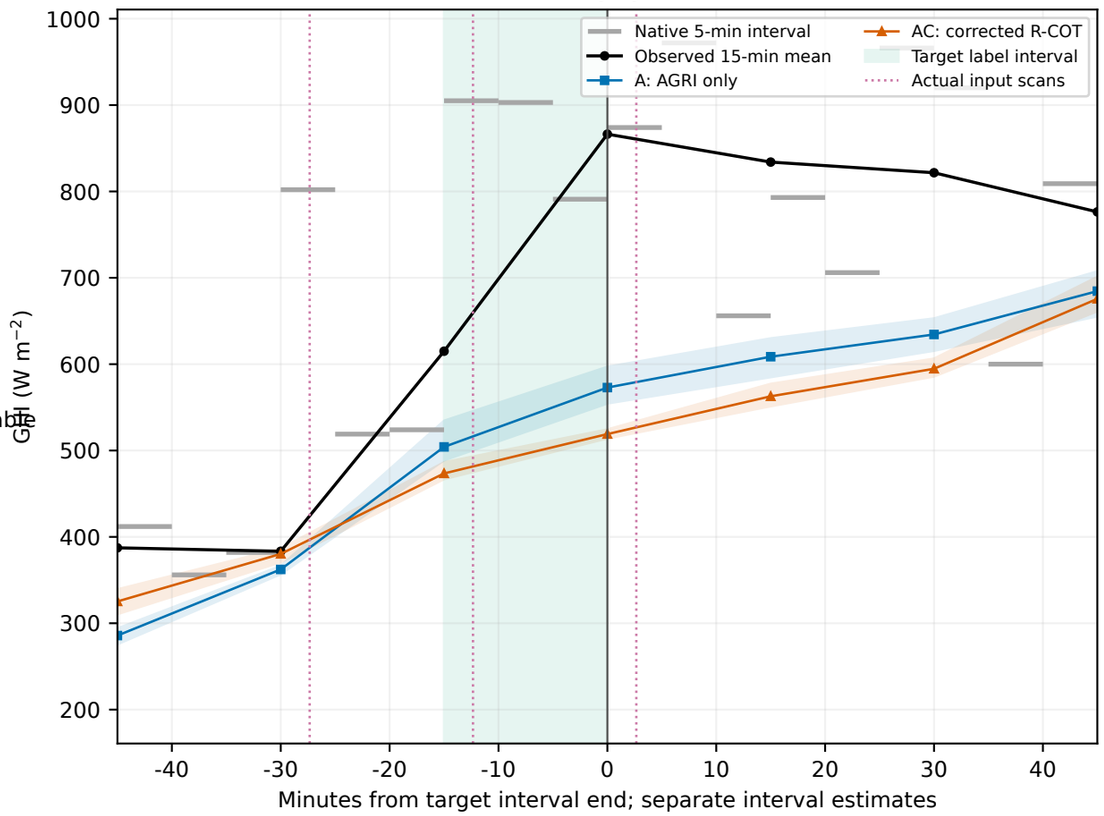
R-COT center 3.63; CPP unavailable



R-COT center 2.60; CPP unavailable



Failure: A low, AC lower in all seeds
2024-05-29 10:45:00 BJT; GHI 866.3, A 572.9, AC 519.0
kt=1.02; local COT sensitivity by frame: -26.7, -30.1, -30.8



Zhujia -30 min
scan 13:17:39 BJT



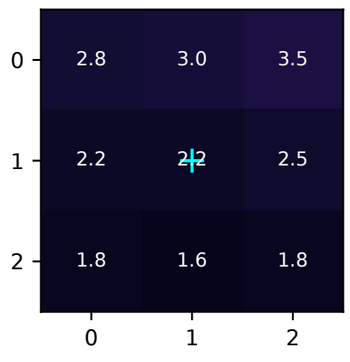
Zhujia -15 min
scan 13:32:39 BJT



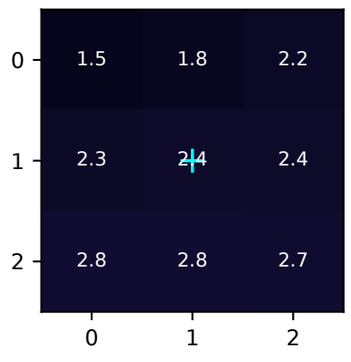
Zhujia +0 min
scan 13:47:39 BJT



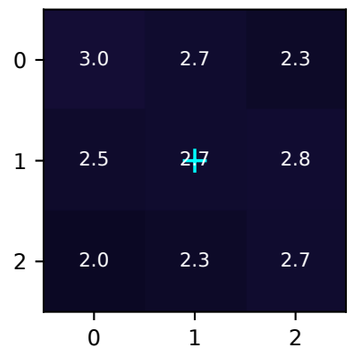
R-COT center 2.16; CPP unavailable



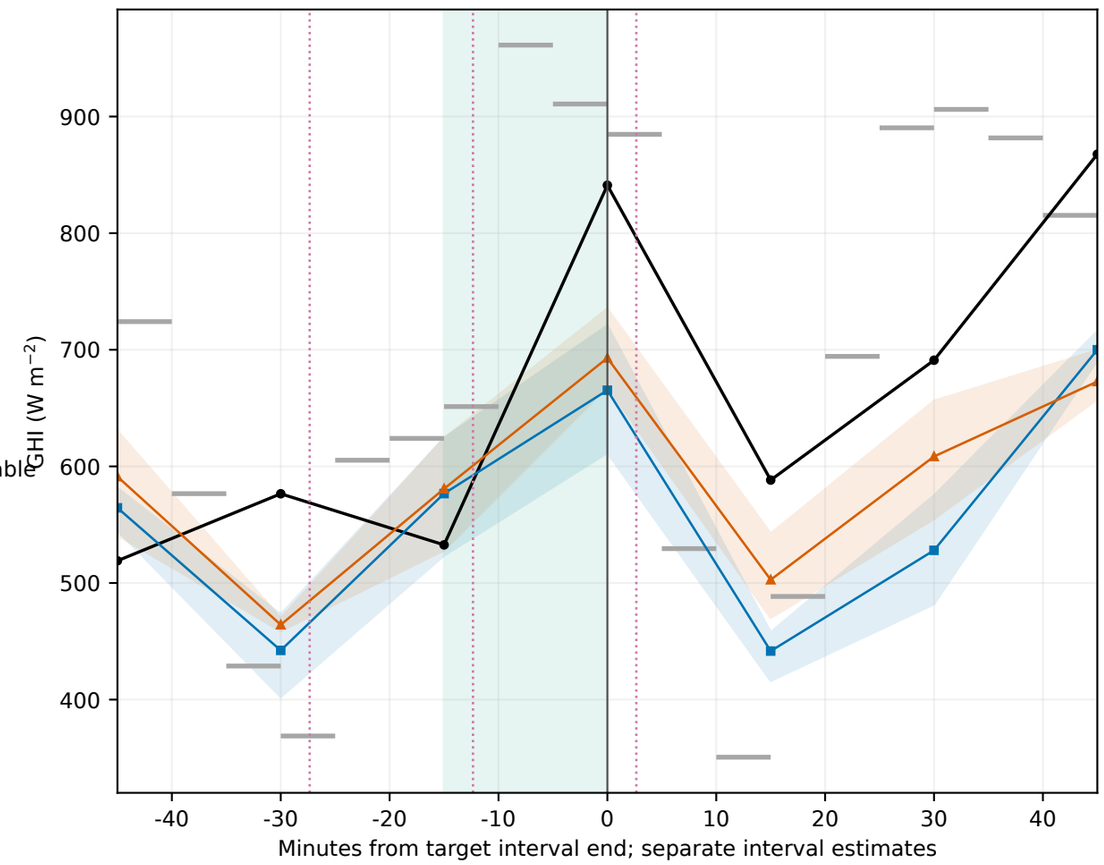
R-COT center 2.35; CPP unavailable



R-COT center 2.68; CPP unavailable



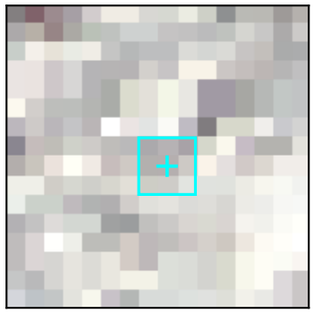
Matched control: A low, AC improves in all seeds
2026-05-06 13:45:00 BJT; GHI 841.1, A 665.3, AC 693.0
kt=0.93; local COT sensitivity by frame: -42.7, -49.4, -50.0



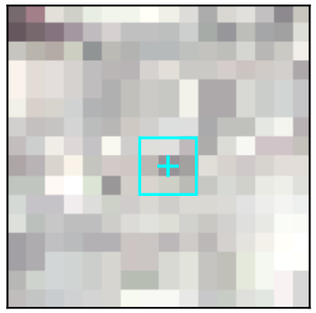
RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

Paired true-cloud forensics 5/6 · fixed 3×3, three-frame GHI models

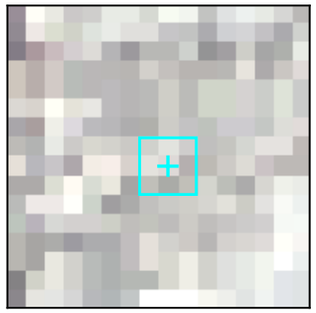
Zhujia -30 min
scan 13:02:39 BJT



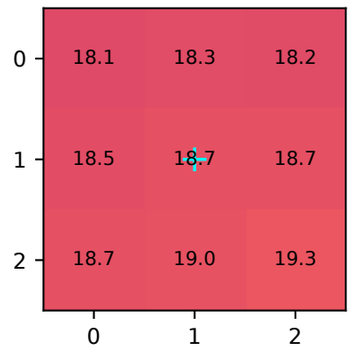
Zhujia -15 min
scan 13:17:39 BJT



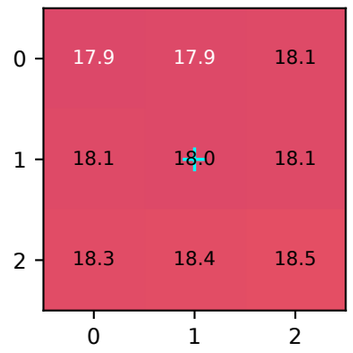
Zhujia +0 min
scan 13:32:39 BJT



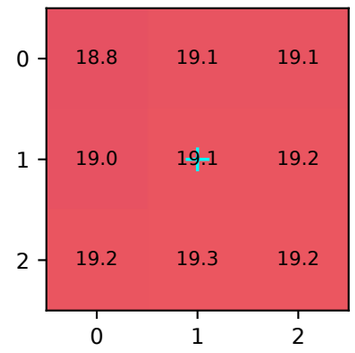
R-COT center 18.66; CPP unavailable



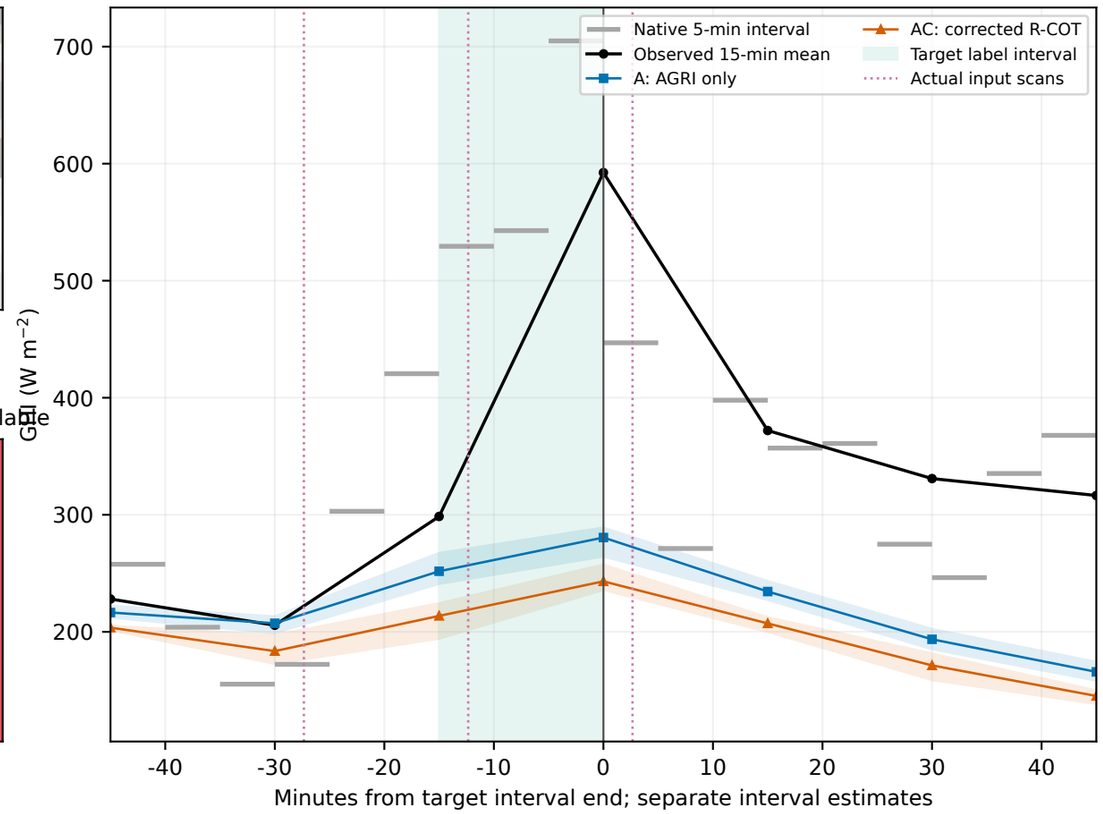
R-COT center 18.01; CPP unavailable



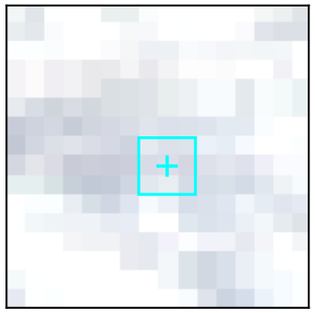
R-COT center 19.15; CPP unavailable



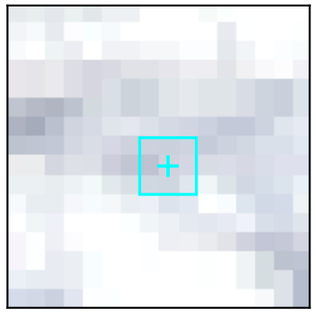
Failure: A low, AC lower in all seeds
2026-06-24 13:30:00 BJT; GHI 592.4, A 280.6, AC 243.1
kt=0.64; local COT sensitivity by frame: -62.5, -69.5, -60.8



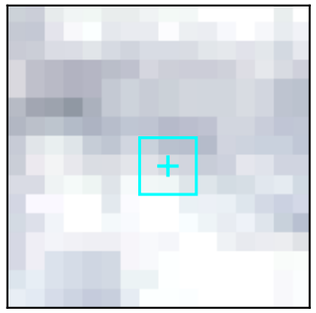
Zhujia -30 min
scan 12:47:39 BJT



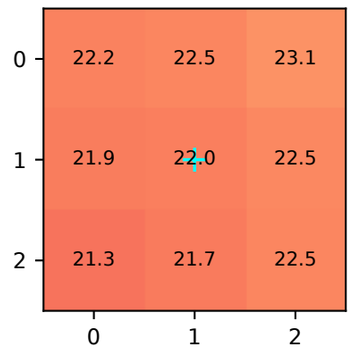
Zhujia -15 min
scan 13:02:39 BJT



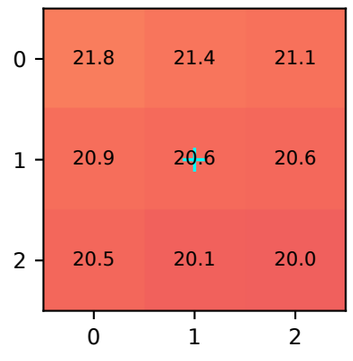
Zhujia +0 min
scan 13:17:39 BJT



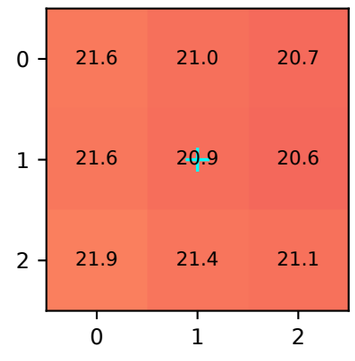
R-COT center 22.02; CPP unavailable



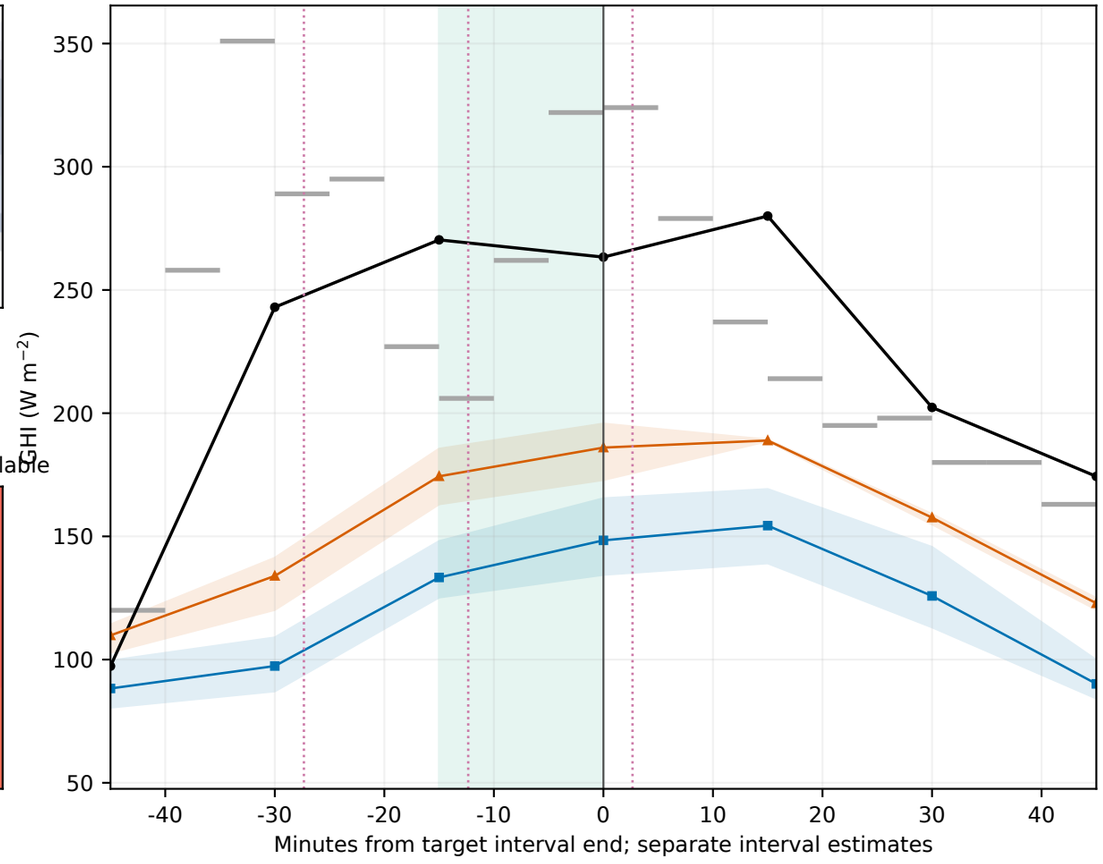
R-COT center 20.63; CPP 46.87

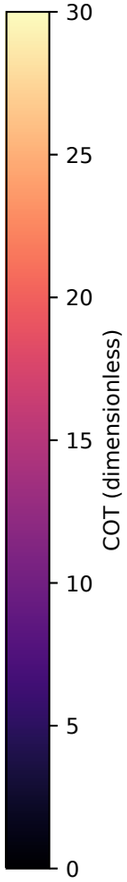


R-COT center 20.88; CPP unavailable



Matched control: A low, AC improves in all seeds
2024-05-30 13:15:00 BJT; GHI 263.3, A 148.4, AC 186.0
kt=0.28; local COT sensitivity by frame: -62.0, -66.5, -60.0

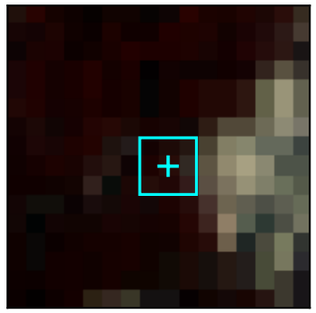




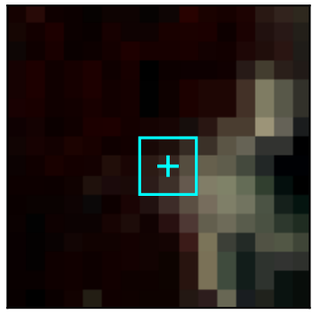
RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

Paired true-cloud forensics 6/6 · fixed 3×3, three-frame GHI models

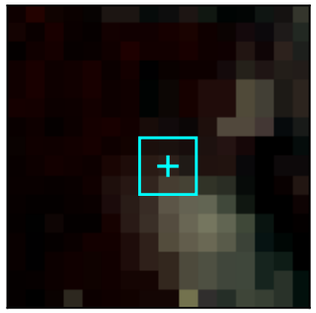
Zhujia -30 min
scan 16:02:39 BJT



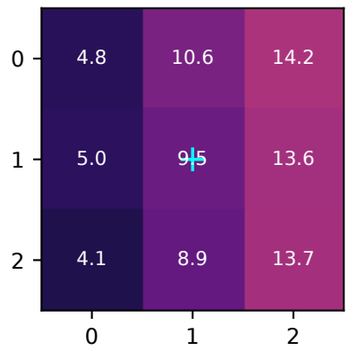
Zhujia -15 min
scan 16:17:39 BJT



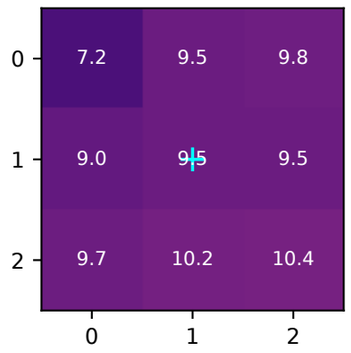
Zhujia +0 min
scan 16:32:39 BJT



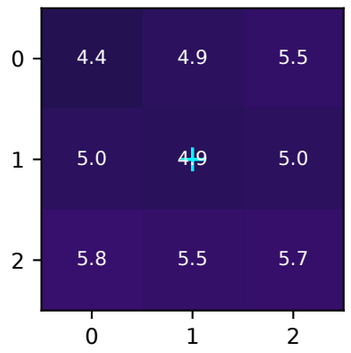
R-COT center 9.50; CPP 22.80



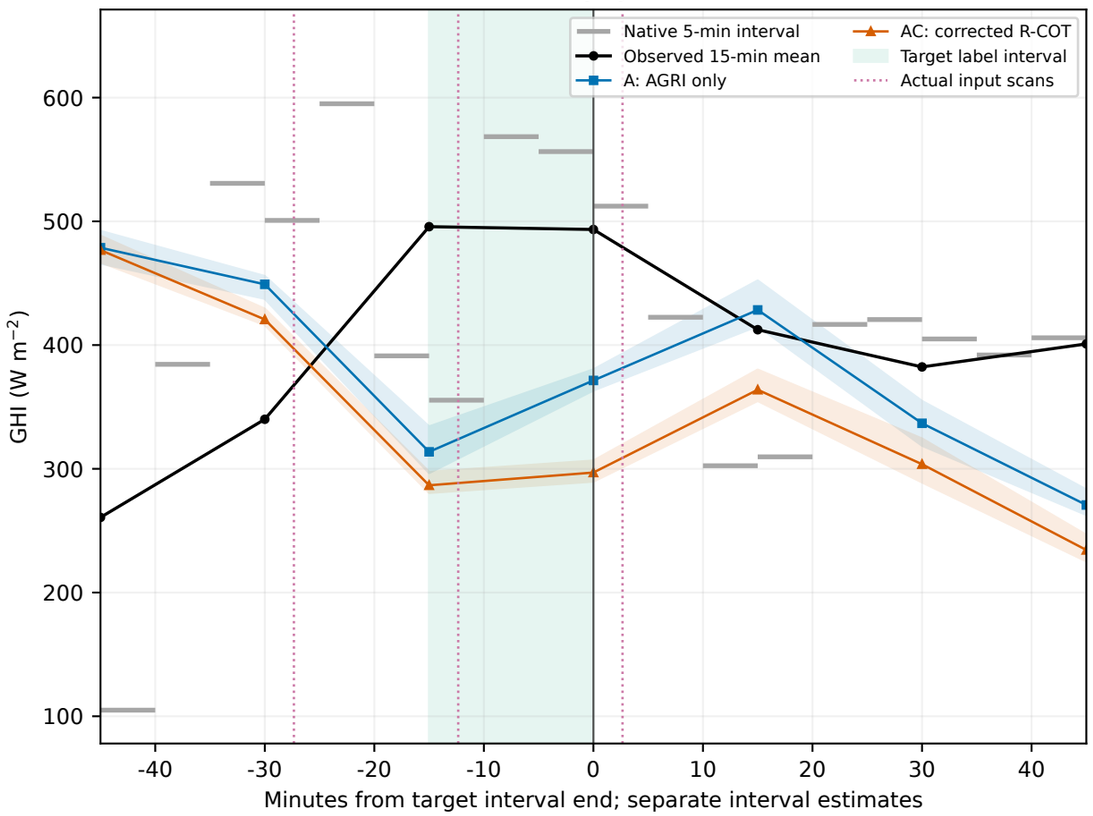
R-COT center 9.54; CPP 13.27

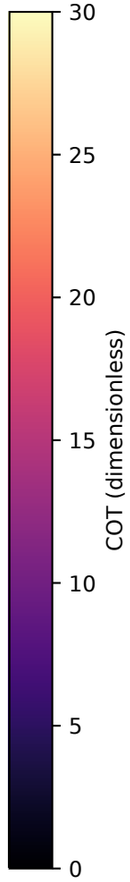


R-COT center 4.87; CPP 5.77



Failure: A low, AC lower in all seeds
2025-07-09 16:30:00 BJT; GHI 493.4, A 371.4, AC 297.0
kt=0.92; local COT sensitivity by frame: -31.7, -35.0, -34.3





Zhujia -30 min
scan 08:17:39 BJT



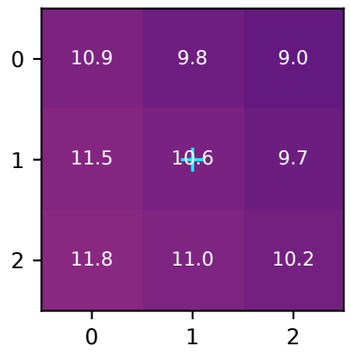
Zhujia -15 min
scan 08:32:39 BJT



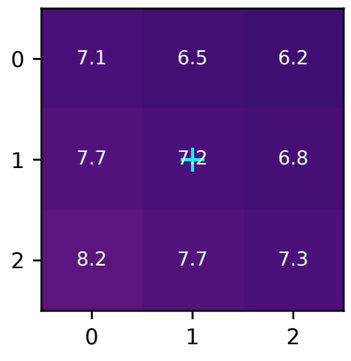
Zhujia +0 min
scan 08:47:39 BJT



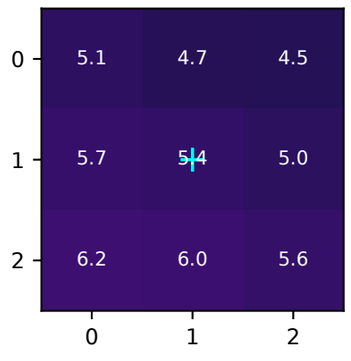
R-COT center 10.56; CPP unavailable



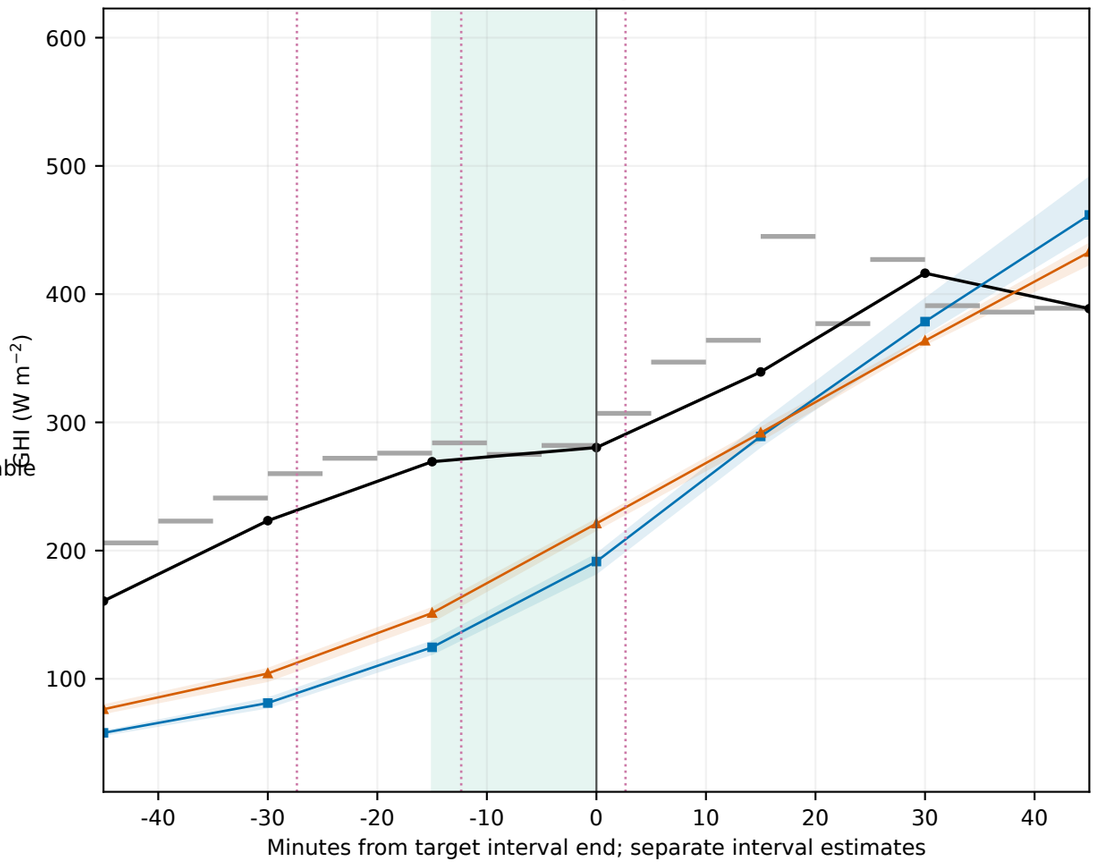
R-COT center 7.23; CPP unavailable



R-COT center 5.41; CPP unavailable



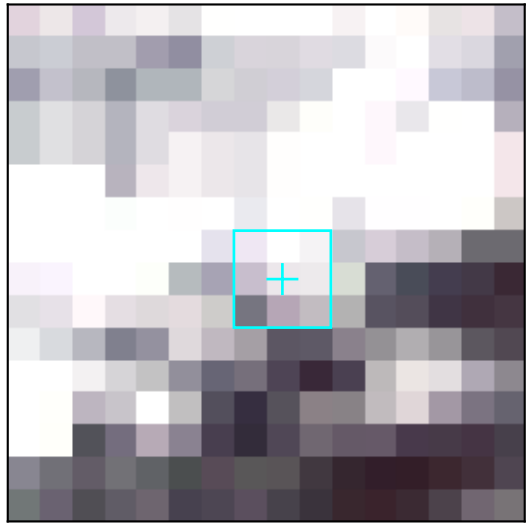
Matched control: A low, AC improves in all seeds
2024-07-09 08:45:00 BJT; GHI 280.3, A 191.5, AC 221.1
kt=0.56; local COT sensitivity by frame: -32.3, -34.3, -31.9



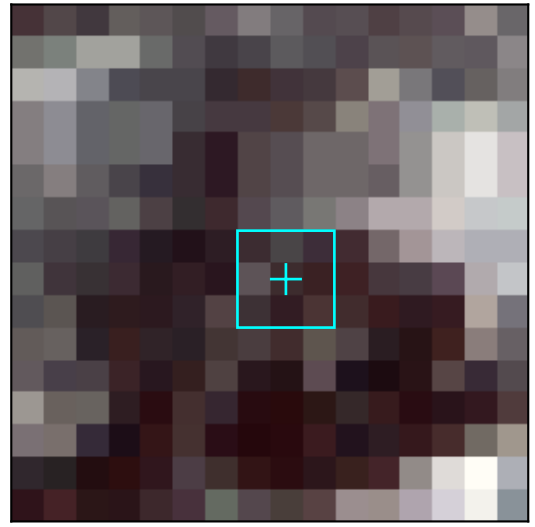
RGB: real C03/C02/C01, shared linear 2–98% stretch. RGB 16×16 is existing R input context only; cyan square is the unchanged GHI 3×3.
COT: exact saved 3×3 input, shared scale. Shaded bands: three-seed range. Five-minute records diagnose timing; metrics remain 15-minute. Local derivatives are model sensitivities.

Existing target-time CTH geometry joined to the current three-frame models

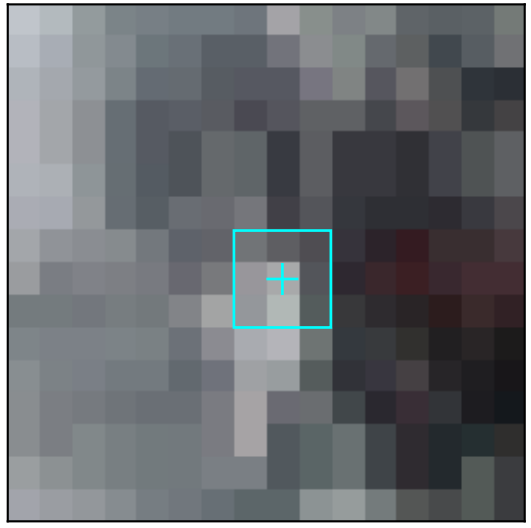
sili 2024-04-19 12:15:00 BJT
GHI 608; current A 340, AC 386
Nearest projected shadow: 6.43 km; helpful seeds 3/3



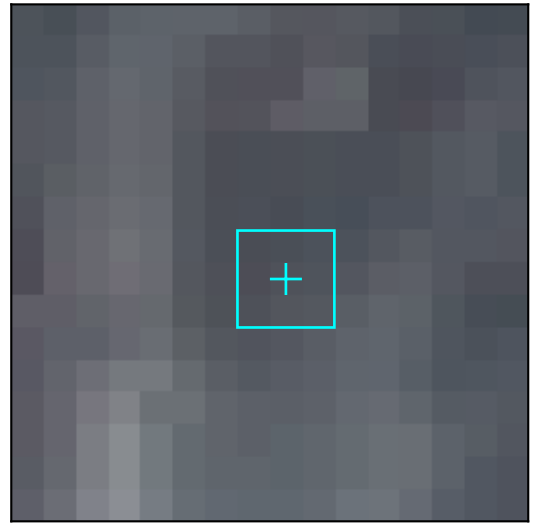
sili 2024-06-11 14:30:00 BJT
GHI 809; current A 569, AC 571
Nearest projected shadow: 1.85 km; helpful seeds 1/3



zhujia 2024-05-29 11:45:00 BJT
GHI 887; current A 604, AC 652
Nearest projected shadow: 2.44 km; helpful seeds 3/3



zhujia 2024-04-20 14:30:00 BJT
GHI 190; current A 502, AC 521
Nearest projected shadow: 1.65 km; helpful seeds 0/3



CTH geometry is reused only at identical validation station/time/grid keys. RGB is qualitative context; 4 km is a product-scale geometric screen, not measured shadow truth.